

VC Formal Tcl Library: Regression Module

Introduction

This list of procedures is designed to help process the regression results to determine whether the current test triggers the failing condition. The failing condition and the messages are customizable to fit into your regression flow. This way, you don't need to post process the log file from the tool report. The list of procedures also contains features to save goals to a file for review and waive for future runs.

Library Location

All files for this lab are located in:
\$VC_STATIC_HOME/auxx/ctg/tcl/library

Directory Structure	
proc/vcf_library_regression.tcl	Tcl procedures related to regression tests
doc/README_regression.pdf	Description of Tcl procedures, usage, and examples for regressions

vcf_library_regression.tcl

This tcl library contains the following user Tcl procedures:

1. save_prop_waiver
2. fpv_rg
3. save_coi_waiver
4. fpv_coi_cov_rg

To use the procedures for your regression test, put this in your tcl file:

```
source $VC_STATIC_HOME/auxx/ctg/tcl/library/proc/vcf_library_regression.tcl
```

Procedure description

Description		Check if any property status falls into regression failure list. It prints out a list of properties from the run with user specified status and save to a file for review as waiver.				
PROCEDURE		No. args	No. defaults			
save_prop_waiver		2	2			
	arguments	waiver_file	prop_status			
	default args	waiver_prop.txt	falsified			
	usage example	save_prop_waiver prop_waiver.txt "falsified uncoverable"				
	expected output	traffic.chk.assert_green_no_waiting_first				

Description		Check if any property status falls into regression failure list. The regression file can contain customized fail or pass message. Waiver file can be provided to compare.				
PROCEDURE		No. args	No. defaults			
fpv_rg		5	5			
	arguments	reg_out_file	prop_status	fail_msg	pass_msg	waiver_file
	default args	sim.out	"falsified vacuous not_run uncoverable"	"ERROR: FORMAL FAIL"	""	""
	usage example	fpv_rg reg.out "falsified not_run" MY_FAIL MY_PASS waiver_prop.txt				
	expected output	ERROR : [FORMAL] 9 properties FAILED and no waiver file				
		ERROR : [FORMAL] traffic.chk.cov_both_green not match any waiver items, FAIL!!!				
		INFO : [FORMAL] all failed items item matched with waivers, PASS!!!				
		INFO: [FORMAL] All properties PASSED, no waiver file				

Description		Check if any coi falls into regression failure list. It prints out a list of registers from the run with user specified status and save to a file for review as waiver.		
PROCEDURE		No. args	No. defaults	
save_coi_waiver		2	2	
	arguments	waiver_file	coi_status	
	default args	waiver_coi.txt	uncovered	
	usage example	save_coi_waiver	coi_waiver.txt "uncovered"	
	expected output	main.outside_reg		

Description		Check if any coi status falls into regression failure list. Message will be printed out to regression log. Waiver file can be provided to compare.				
PROCEDURE		No. args	No. defaults			
fpv_coi_cov_rg		3	3			
	arguments	reg_out_file	coi_status	waiver_file		
	default args	sim.out	uncovered	waiver_coi.txt		
	usage example	fpv_coi_cov_rg reg.out uncovered waiver_coi.txt				
	expected output	INFO : [FORMAL] Assertion COI seq coverage uncovered matched with waivers, PASS!!! ERROR : [FORMAL] Seq first.outside_reg uncovered does not match any waiver items, FAIL!!!				